Midterm Examination

① $a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{2\pi}\int_0^{2\pi} \left(\frac{\pi - x}{2}\right) dx = 0$ ——— (1)

$a_n = \frac{1}{\pi}\int_0^{2\pi} \frac{(\pi - x)}{2}\cos nx\,dx = 0$ ——— (1)

$b_n = \frac{1}{\pi}\int_0^{2\pi} \left(\frac{\pi - x}{2}\right) dx = 1/n$ ——— (1)

$f(x) = \sum_{n=1}^{\infty} \frac{1}{n}\sin nx$

Put $x = \pi/2$ and get $1 - 1/3 + 1/5 - 1/7 + \cdots = \pi/4$ ——— (1)

② $f(t) = \frac{7t - 9}{t(t-1)(t-3)}$ where $t = x + 1$

$= \frac{A}{t} + \frac{B}{(t-1)} + \frac{C}{(t-3)}$

on solving we get $A = -3$, $B = 1$, $C = 2$

$f(t) = \frac{-3}{t} + \frac{1}{(t-1)} + \frac{2}{(t-3)}$ ——— (1)

i) when $|t| < 1$ ie $|x+1| < 1$

$f(t) = \frac{-3}{t} + \frac{1}{(-1)(1-t)} + \frac{2}{(-3)(1-t/3)}$

$= \frac{-3}{t} - \left[1 + t + t^2 + \cdots\right] - \frac{2}{3}\left[1 + t/3 + (t/3)^2 + \cdots\right]$ —— (1)

ii) when $1 < |t| < 3$ ie $1 < |x+1| < 3$

$f(t) = \frac{-3}{t} + \frac{1}{t(1 - 1/t)} + \frac{2}{(-3)(1 - t/3)}$

$$= \frac{-3}{t} + \frac{1}{t}\left[1 + \frac{1}{t} + \frac{1}{t^2} + \frac{1}{t^3} + \cdots\right] - \frac{2}{3}\left[1 + \frac{t}{3} + \left(\frac{t}{3}\right)^2 + \cdots\right]$$

(1)

when $|t| > 3$ ie $|z+1| > 3$

$$f(t) = \frac{-3}{t} + \frac{1}{t(1-1/t)} + \frac{2}{t(1-3/t)}$$

$$= \frac{-3}{t} + \frac{1}{t}\left[1 + \frac{1}{t} + \frac{1}{t^2} + \cdots\right] + \frac{2}{t}\left[1 + \frac{3}{t} + \left(\frac{3}{t}\right)^2 + \cdots\right]$$

(1)

Three Marks

(3) $\int_C \frac{4 - 3z}{z(z-1)(z-2)}\, dz$ where $C: |z-1| = 2$

Here $z = 1, 0, 2$ are the simple poles and all three lie inside & on 'C'

$$\therefore \int f(z)\, dz = 2\pi i \left[\operatorname*{res}_{z=0} f(z) + \operatorname*{res}_{z=1} f(2) + \operatorname*{res}_{z=2} f(2)\right]$$

$$\operatorname*{res}_{z=0} f(z) = \lim_{z \to 0} \frac{4 - 3z}{(z-1)(z-2)} = 2$$

$$\operatorname*{res}_{z=1} f(z) = \lim_{z \to 1} \frac{4 - 3z}{z(z-2)} = -1$$

$$\operatorname*{res}_{z=2} f(z) = \lim_{z \to 2} \frac{4 - 3z}{z(z-1)} = -1$$

$$\therefore \int f(z)\, dz = 2\pi i \left[2 - 1 - 1\right] = 0$$

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(4) $$b_n = \frac{4}{T}\int_0^{T/2} f(x)\sin\frac{2\pi nx}{T}\,dx$$

$T = 2\cdot 1$

$$\therefore b_n = 2\int_0^1 \cos\pi x \sin n\pi x\,dx$$

$$= \int_0^1 \{\sin(1+n)\pi x - \sin(1-n)\pi x\}dx$$

$$= \frac{2n[(-1)^n+1]}{\pi(n^2-1)}, \quad n\neq 1 \qquad \text{——} \frac{1}{2}$$

$$b_1 = 2\int_0^1 \cos\pi x \sin\pi x\,dx = 0 \qquad \text{——} 01$$

$$\therefore f(x) = \sum_{n=2}^{\infty} b_n \sin n\pi x \qquad \text{——} 0\frac{1}{2}$$

$$= \frac{2}{\pi}\sum_{n=2}^{\infty}\frac{n[(-1)^n+1]}{(n^2-1)}\sin n\pi x \qquad \underline{03}$$

(5) Let $f = \nabla\cdot\vec{u} = 4(x^3+y^3+z^3)$

$$\nabla f|_{(4,4,2)} = 12(x^2i+y^2j+z^2k)|_{(4,4,2)}$$

$$= 48(4i+4j+k) \qquad \text{——} \frac{1}{2}$$

Normal to the sphere $S: x^2+y^2+z^2=36$ is

$$\nabla S = 2(xi+yj+zk)|_{(4,4,2)} \qquad \text{——} \frac{1}{2}$$

$$= 4(2i+2j+2k)$$

$$\hat{a} = \frac{\nabla S}{|\nabla S|} = \frac{2i+2j+k}{3} \qquad \text{——} \frac{1}{2}$$

(a)

$\therefore \nabla f \cdot \hat{a} = 272.$ — — — (½)

(b) $u = x^2 - y^2 - y$

$\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y}$

$v(x,y) = 2xy + g(x)$ — — — (½)

$\frac{\partial v}{\partial x} = 2y + g'(x) = -\frac{\partial u}{\partial y}$

$\therefore 2y + g'(x) = 2y + 1 \Rightarrow g'(x) = 1$ — (½)

$\Rightarrow g(x) = x$

$\therefore v(x,y) = 2xy + x.$ — (½)

$\therefore f(z) = u + iv = x^2 - y^2 - y + i(2xy + x)$ — (½)

Put $x = z, \; y = 0$

$f(z) = z^2 + iz.$

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(c) $f(x) = xe^{-x}, \quad 0 < x < \infty.$

$F(s) = \frac{1}{\sqrt{2\pi}} \int_0^\infty x e^{-x} e^{isx} dx$ — — — (½)

$= \frac{1}{\sqrt{2\pi}} \int_0^\infty x e^{-(1+is)x} dx.$

$= \frac{1}{\sqrt{2\pi}} \left[x \frac{e^{-(1+is)x}}{-(1+is)} - \frac{e^{-(1+is)x}}{(1+is)^2} \right]_0^\infty$ — (½)

$= \frac{1}{\sqrt{2\pi}(1+is)^2}$

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